

```

#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include <unistd.h>
#include <ctype.h>
#include <windows.h>
#include <process.h>
#include <unistd.h>

#define UP 72
#define DOWN 80
#define LEFT 75
#define RIGHT 77
#define STARTLENGTH 5

COORD ncoord;

int SNAKESTARTX=20;
int SNAKESTARTY=25;

int XBORDER;
int YBORDER;
int length=STARTLENGTH;;
int bend_no;
int len;
char key;
int life=3;
int level;
char speed;

void Print_welcome();
void Start();
void GotoXY(int x,int y);
void Move();
void Right();
void Left();
void Down();
void Up();
void Bend();
void Erase();
void Loses_Management();
void Food();
int Score();
void Delay();
char Menu();
void Print_maze(int slevel);
void record();

struct coordinate
{
    int x;
    int y;
    int direction;
};

typedef struct coordinate coordinate;

coordinate head, bend[500], food, poison[4], body[30];

int main()
{
    Print_welcome();
    Start();

    return 0;
}

```

```

}

void Move()
{
    int a,i;
    Score();

    if (length >= 30)
    {
        level++;
        Start();
    }

    do
    {
        fflush(stdin);
        len=0;

        if (speed == 'E' || speed == 'e')
        {
            Delay();
            Delay();
            Delay();
        }
        else if (speed == 'H' || speed == 'h')
            Delay();

        Erase();

        if(head.direction==RIGHT)
        {
            ncoord.X = head.x + 1;
            ncoord.Y = head.y;
            Right();
        }
        else if(head.direction==LEFT)
        {
            ncoord.X = head.x - 1;
            ncoord.Y = head.y;
            Left();
        }
        else if(head.direction==DOWN)
        {
            ncoord.X = head.x;
            ncoord.Y = head.y + 1;
            Down();
        }
        else if(head.direction==UP)
        {
            ncoord.X = head.x;
            ncoord.Y = head.y - 1;
            Up();
        }

        Loses_Management();

    }
    while(!kbhit());

    a=getch();

    if ( a == 'M' || a == 'm')
        Menu();

    key=getch();

    if((key==RIGHT&&head.direction!=LEFT&&head.direction!=RIGHT)||
    (key==LEFT&&head.direction!=RIGHT&&head.direction!=LEFT)||
    (key==UP&&head.direction!=DOWN&&head.direction!=UP)||
    (key==DOWN&&head.direction!=UP&&head.direction!=DOWN))

```

```

{
    bend_no++;

    bend[bend_no]=head;

    head.direction=key;

    Move();

}
else
{
    printf("\a");
    Move();
}
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
void GotoXY(int x, int y)
{
    HANDLE a;
    COORD b;
    fflush(stdout);
    b.X = x;
    b.Y = y;
    a = GetStdHandle(STD_OUTPUT_HANDLE);
    SetConsoleCursorPosition(a,b);
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
void Down()
{
    int i;
    for(i=0; i<=(head.y-bend[bend_no].y)&&len<length; i++)
    {
        GotoXY(head.x, head.y-i);
        {
            if(len==0)
                printf("v");
            else
                printf("*");
        }
        body[len].x=head.x;
        body[len].y=head.y-i;
        len++;
    }
    Bend();

    head.y++;
}
void Delay()
{
    long double i;
    for(i=0; i<=(10000000); i++);
}
void Loses_Management()
{
    char a;
    int reti;

    HANDLE display=GetStdHandle(STD_OUTPUT_HANDLE);

    ReadConsoleOutputCharacter(display,&a,1,ncoord,&reti);

```

```

if(a==35)
{
    if(life <= 0)
        goto ENDGAME;
    life--;
}

else if(a==64)
{
    length++;
    Food();
}

else if(a!=32 && a!= 64)
{
    life--;
    if(life >= 0)
    {
        Print_maze(level);
        Move();
    }
    else
    {
        ENDGAME:
        system("cls");
        printf("All lives completed\nBetter Luck Next Time!!!\nPress any key to quit
the game\n");
        record();
        exit(0);
    }
}

}

void Food()
{
    char a;
    int reti;

    COORD fcoord;
    HANDLE display=GetStdHandle(STD_OUTPUT_HANDLE);

    food.x=rand()% XBORDER;
    if(food.x==0)
        food.x=1;
    food.y=rand()% YBORDER;
    if(food.y<=10)
        food.y+=11;

    ReadConsoleOutputCharacter(display,&a,1,fcoord,&reti);

    while (a!=32)
    {
        if(food.x>=XBORDER/2)
            food.x--;
        else
            food.x++;

        if(food.y>=YBORDER/2)
            food.y--;
        else
            food.y++;

        fcoord.X=food.x;
        fcoord.Y=food.y;

        ReadConsoleOutputCharacter(display,&a,1,fcoord,&reti);
    }
}

```

```

    }

    GotoXY(food.x, food.y);

    printf("%c", 64);
}
void Left()
{
    int i;
    for(i=0; i<=(bend[bend_no].x-head.x)&&len<length; i++)
    {
        GotoXY((head.x+i), head.y);
        {
            if(len==0)
                printf("<");
            else
                printf("*");
        }
        body[len].x=head.x+i;
        body[len].y=head.y;
        len++;
    }
    Bend();

    head.x--;
}
void Right()
{
    int i;
    for(i=0; i<=(head.x-bend[bend_no].x)&&len<length; i++)
    {
        body[len].x=head.x-i;
        body[len].y=head.y;
        GotoXY(body[len].x, body[len].y);
        {
            if(len==0)
                printf(">");
            else
                printf("*");
        }

        len++;
    }
    Bend();
    head.x++;
}
void Bend()
{
    int i, j, diff;
    for(i=bend_no; i>=0&&len<length; i--)
    {
        if(bend[i].x==bend[i-1].x)
        {
            diff=bend[i].y-bend[i-1].y;
            if(diff<0)
                for(j=1; j<=(-diff); j++)
                {
                    body[len].x=bend[i].x;
                    body[len].y=bend[i].y+j;
                    GotoXY(body[len].x, body[len].y);
                    printf("*");
                    len++;
                    if(len==length)
                        break;
                }
        }
    }
}

```

```

        else if(diff>0)
        {
            for(j=1; j<=diff; j++)
            {
                body[len].x=bend[i].x;
                body[len].y=bend[i].y-j;
                GotoXY(body[len].x, body[len].y);
                printf("*");
                len++;
                if(len==length)
                    break;
            }
        }
        else if(bend[i].y==bend[i-1].y)
        {
            diff=bend[i].x-bend[i-1].x;
            if(diff<0)
            {
                for(j=1; j<=(-diff)&&len<length; j++)
                {
                    body[len].x=bend[i].x+j;
                    body[len].y=bend[i].y;
                    GotoXY(body[len].x, body[len].y);
                    printf("*");
                    len++;
                    if(len==length)
                        break;
                }
            }
            else if(diff>0)
            {
                for(j=1; j<=diff&&len<length; j++)
                {
                    body[len].x=bend[i].x-j;
                    body[len].y=bend[i].y;
                    GotoXY(body[len].x, body[len].y);
                    printf("*");
                    len++;
                    if(len==length)
                        break;
                }
            }
        }
    }
}

void Print_welcome()
{
    GotoXY(10,10);
    printf("\tWelcome to the mini Snake game.(press any key to continue)\n");
    getch();
    system("cls");
    GotoXY(10,10);
    printf("\tGame instructions:\n");
    printf("\n-> Use arrow keys to move the snake.\n");
    printf("\n-> Everytime you eat food (@) the length of the snake will be increased by 1 element and thus the score.\n");
    printf("\n-> Here you are provided with three lives. Your life will decrease as you hit the wall or snake's body or poison (#).\n");
    printf("\n-> You can pause the game in the middle by pressing any key. To continue the paused game press any other key once again.\n");
    printf("\n\nPress any key to play the game...");
    getch();
}

void record()
{
    char pname[20], nplname[20], cha, c;
    int i, j, px;
    FILE *info;
    info=fopen("record.txt", "a+");
    getch();
    system("cls");

```

```

printf("Enter your name\n");
scanf("%[^\n]", plname);
//*****
for(j=0; plname[j]!='\0'; j++)
{
    nplname[0]=toupper(plname[0]);
    if(plname[j-1]==' ')
    {
        nplname[j]=toupper(plname[j]);
        nplname[j-1]=plname[j-1];
    }
    else nplname[j]=plname[j];
}
nplname[j]='\0';
//*****

fprintf(info, "Player Name :%s\n", nplname);

time_t mytime;
mytime = time(NULL);
fprintf(info, "Played Date:%s", ctime(&mytime));
//*****

fprintf(info, "Score:%d\n", px=Score());

system("cls");

for(i=0; i<=50; i++)
    fprintf(info, "%c", '_');
fprintf(info, "\n");
fclose(info);

printf("Wanna see past records press 'y'\n");

cha=getch();
system("cls");
if(ch=='y')
{
    info=fopen("record.txt", "r");
    do
    {
        putchar(c=getc(info));
    }
    while(c!=EOF);
}
fclose(info);

cha=getch();
}
int Score()
{
    int score;
    GotoXY(20, 8);
    score=length-5;
    printf("SCORE : %d", score);
    GotoXY(50, 8);
    printf("Life : %d", life);
    return score;
}
void Up()
{
    int i;
    for(i=0; i<=(bend[bend_no].y-head.y)&&len<length; i++)
    {

```

```

        GotoXY(head.x, head.y+i);
        {
            if(len==0)
                printf("^");
            else
                printf("*");
        }
        body[len].x=head.x;
        body[len].y=head.y+i;
        len++;
    }
    Bend();
    head.y--;
}
void Erase()
{
    int i, j;

    for(i=0; i<=length; i++)
    {
        GotoXY(body[i].x, body[i].y);
        printf(" ");
    }
}

void Start()
{
    system("cls");
    if (level == 0)
    {
        GotoXY(36,14);
        printf("Choose the level of hardness");
        GotoXY(36,15);
        printf("Easy 'E' \t\t Hard 'H' ");
        GotoXY(36,16);
        speed=0;
        while (!(speed == 'e' || speed == 'E' || speed == 'h' || speed == 'H'))
            speed = getch();
    }

    system ("cls");
    if (level == 3)
    {
        GotoXY(10,10);
        printf("Congratulations!!!");
        GotoXY(10,12);
        printf("You have won the game!");
        record();
        exit(0);
    }
    GotoXY(36,14);
    printf(" Welcome to level %d", level+1);
    printf("\n\t\t Press any key to play the game...");
    getch();

    system ("cls");
    Print_maze(level);

    Move();
}

char Menu()
{
    int i, j;

```

```

char a;

GotoXY(90,16);
printf ("Menu");
GotoXY(90,17);
printf ("press the corresponding letter to each");
GotoXY(90,18);
printf ("option to select it");
GotoXY(90,19);
printf ("Start a new game:  'N'");
GotoXY(90,20);
printf ("Exit:      'E'");
GotoXY(90,21);
printf ("Exit Menu:  'C'");
GotoXY(90,22);

a = getch();

//For starting a new game
if ( a == 78 || a == 110)
{
    GotoXY(90,23);
    printf ("Are you sure you want to ");
    GotoXY(90,24);
    printf ("start a new game? ");
    GotoXY(90,25);
    printf ("Yes : 'Y'      No : 'N'");
    a = getch();

    if ( a == 78 || a == 110)
    {
        flag:
        GotoXY(90,23);
        for (i = 90; i < 128 ; i++ )
        {
            for (j = 22; j < 26; j++)
            {
                GotoXY(i,j);
                printf(" ");
            }
        }
        Menu();
    }

    else if ( a == 89 || a == 121)
    {
        level=0;
        Start();
    }
}
else goto flag;
}

//for erasing the menu
else if (a == 99 || a == 67)
{
    GotoXY(90,16);
    for (i = 90; i < 131 ; i++ )
    {
        for (j = 16; j < 30; j++)
        {
            GotoXY(i,j);
            printf(" ");
        }
    }
    return ;
}

```

[illegible]

[illegible]